SPA617 — RF Components: Characterization and Measurement

Experiment: Characterization of a Notch Filter

Objective

To characterize the transmission characteristics of a notch filter by measuring the output power over a range of frequencies and identifying the notch (centre) frequency and attenuation.

Components Required

Notch RF filter, RF signal source, spectrum analyzer/VNA, RF cables and connectors, and suitable 50 Ω termination/load.

Theory

A notch filter, also called a band-reject filter, strongly attenuates a narrow range of frequencies around its centre frequency while allowing frequencies outside the stopband to pass with comparatively lower attenuation.

The attenuation/insertion loss is calculated as:

Attenuation (dB) = $P_{in} - P_{out}$

Here, P_{in} is the input/reference power measured without the filter and P_{out} is the output power measured with the filter. The notch frequency corresponds approximately to the frequency at which the attenuation is maximum (or output power is minimum).

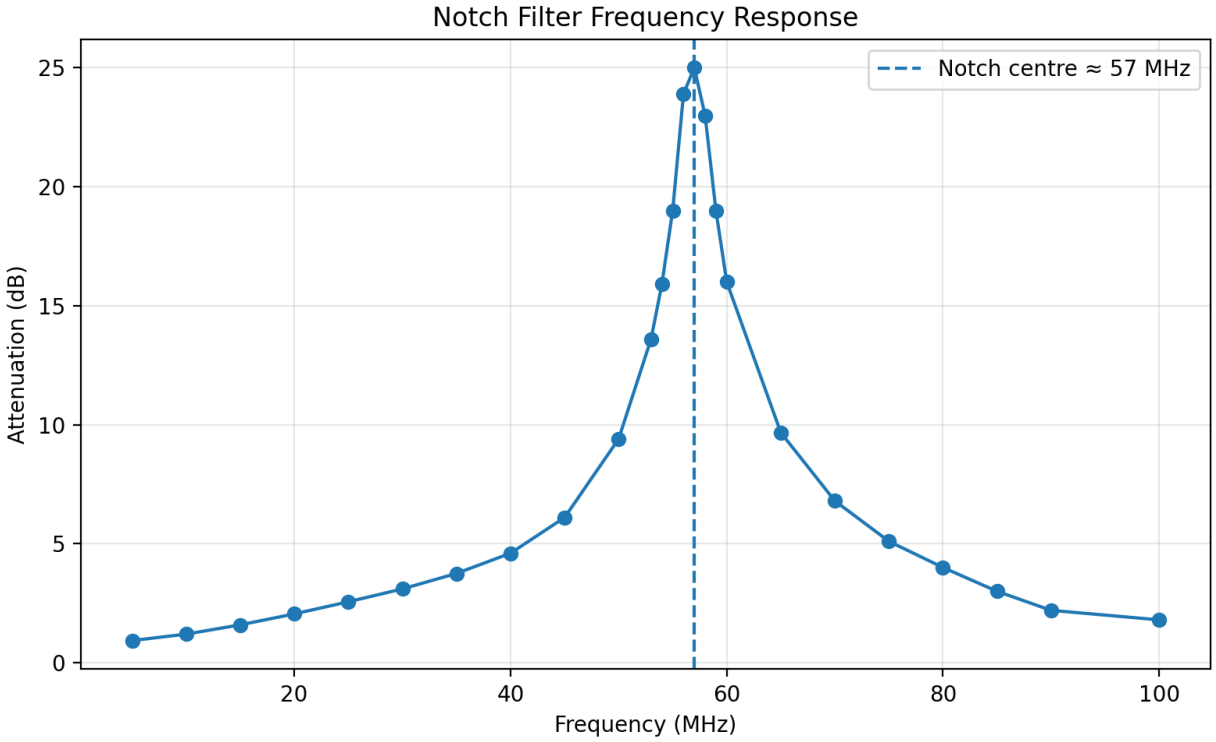
Experimental Setup

The RF source was connected to the measuring instrument through the notch filter. The reference input power was measured without the filter. The filter was then inserted and the output power was measured while the frequency was varied from 5 MHz to 100 MHz. The input power was maintained at 0 dBm.

Observation & Graphs

Input Frequency (MHz)	P_in without filter (dBm)	P_out with filter (dBm)	Attenuation (dB)
5	0.00	-0.93	0.93
10	0.00	-1.20	1.20
15	0.00	-1.59	1.59
20	0.00	-2.05	2.05
25	0.00	-2.56	2.56
30	0.00	-3.10	3.10
35	0.00	-3.75	3.75
40	0.00	-4.60	4.60
45	0.00	-6.10	6.10
50	0.00	-9.40	9.40
53	0.00	-13.60	13.60
54	0.00	-15.90	15.90
55	0.00	-19.00	19.00
56	0.00	-23.90	23.90

Input Frequency (MHz)	P_in without filter (dBm)	P_out with filter (dBm)	Attenuation (dB)
57	0.00	-25.00	25.00
58	0.00	-23.00	23.00
59	0.00	-19.00	19.00
60	0.00	-16.00	16.00
65	0.00	-9.67	9.67
70	0.00	-6.80	6.80
75	0.00	-5.10	5.10
80	0.00	-4.00	4.00
85	0.00	-3.00	3.00
90	0.00	-2.20	2.20
100	0.00	-1.80	1.80



Calculation & Results

The input/reference power was maintained at P_{in} = 0 dBm. Therefore:

Attenuation = P_{in} − P_{out}

At 57 MHz, $P_{out} \approx$ –25.0 dBm. Therefore, attenuation = 0 – (–25.0) = **25.0 dB**.

The maximum attenuation in the measured data occurs at approximately **57 MHz**. Hence, the measured notch centre frequency is approximately **57 MHz**, with a maximum attenuation of about **25 dB**.

Result: The notch filter exhibits a pronounced rejection around 57 MHz and comparatively lower attenuation away from the notch region.

Conclusion / Key Takeaway

The transmission characteristics of the notch filter were successfully measured. The filter strongly suppresses the frequency components around its notch frequency while allowing frequencies away from the rejection region to pass with lower attenuation. From the measured data, the notch centre frequency is approximately 57 MHz and the maximum observed attenuation is about 25 dB.

Note: The observation values used in this report are transcribed from the provided laboratory observation sheet. The value at 57 MHz is read as approximately –25 dBm from the handwritten entry.